# SIDDHI PRAVIN LIPARE

MS by Research · CVIT, IIIT Hyderabad

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## EDUCATION

**MS by Research, Computer Science & Engineering** — Jan 2026 – Present
*International Institute of Information Technology, Hyderabad* — CGPA 8.00

**B.S. Data Science & Engineering** — Dec 2021 – May 2025
*Indian Institute of Science Education & Research, Bhopal* — CGPA 8.41

Senior Secondary (Class XII), CBSE Board · 90.0% — 2021

Secondary (Class X), CBSE Board · 95.0% — 2019

## EXPERIENCE

**Centre for Visual Information & Technology (CVIT), IIIT Hyderabad** — May 2025 – Dec 2025
*Applied Research Fellow* — Offer Letter

- Designed **RoadTones** – a dataset–model–evaluation framework for tone-controllable road-video captioning. Work accepted at **CVPR Findings 2026** (Project page).
- Developed key components of the dataset generation pipeline and evaluation metrics, enabling comprehensive benchmarking against 10 state-of-the-art Video-Language Models (VLMs).
- Investigated and optimized the vLLM inference pipeline for the project demo, minimizing video quality degradation during inference to preserve caption generation performance.

**Indian Institute of Technology Madras** — May 2024 – Jul. 2024
*Research Intern* — Certificate, Report

- Explored Super-Resolution techniques for 3D medical imaging, enhancing z-axis resolution in CT and MRI scans.
- Utilized GANs, Diffusion Models, and NeRF to improve image clarity for better diagnostics.

## PROJECTS

**BS Thesis – Image Tampering Detector** — Jan. 2025 – April 2025
*IISER Bhopal* — Report

- Leveraged the use of Discrete Wavelet Transform along with an Adaptive Sub-Band Attention Module to capture frequency-spatial cues to enable precise localization of tampered regions in an image.

**Improving Monocular Depth Estimation of Outdoor Images in Dim Light Conditions** — Sept. 2024 – Nov. 2024
*IISER Bhopal* — Report

- Integrated denoising and deblurring filters into the transformer-based DepthAnything V2 model to improve depth estimation in blurry and noisy nighttime images.

**Prediction of Mortality Rate of Heart Failure Patients Admitted to ICU** — Sept. 2023 – Nov. 2023
*IISER Bhopal* — GitHub

- A binary classification problem executed using supervised machine-learning techniques. Several classifiers and pre-processing methods were applied to determine which classifier gave the best results.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python3, C/C++, JavaScript, MATLAB, R, HTML, CSS, MySQL |
| **Technologies** | PyTorch, TensorFlow, Keras, OpenCV |
| **Other** | Docker, Firebase, Android SDK, Data Structures & Algorithms, Linux |

## KEY COURSES

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|---|---|
| **Mathematics** | Linear Algebra, Multivariable Calculus, Discrete Mathematics, Probability & Statistics |
| **DS & Engineering** | 3D Deep Learning, Natural Language Processing, Data Analyses & Statistics for Geosciences, Applied & Accelerated AI, Deep Learning, Artificial Intelligence, Machine Learning, Computer Vision, Database Management Systems, Applied Optimisation, Econometrics, Data Science in Practice (Advanced Python), Data Structures & Algorithms, Signals & Systems |

## ACHIEVEMENTS

**Publication** — *'RoadTones: Tone Controllable Text Generation from Road Event Videos'* accepted at the CVPR 2026 Findings Track. — 2026

**Presentation** — Poster *'Improving Image Tampering Detection: A Hybrid Approach Using Frequency and Spatial Domain Features'* presented at CV4Science Workshop, CVPR 2025. — 2025

**ISRO Robotics Challenge** — Among the Top 30 teams out of 400 in India. — 2024

**IEEE CAS Student Design Competition** — 3rd place in the Asia-Pacific Region. — 2022